SIMATS ENGINEERING

ECA09 - DIGITAL SIGNAL PROCESSING LAB

Experiment 06.2

Analysis of Finite Word Length Effects in Digital Signal Representation Using Truncation and Rounding Techniques in MATLAB.

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Aim

To implement and analyze the experiment described in the official ECA09 MATLAB lab manual.

Procedure / Calculations

Follow the experiment workflow given in the official lab manual: initialize MATLAB, define the specified signal/filter parameters, execute the algorithm, plot the required responses, and record the result.

MATLAB Program

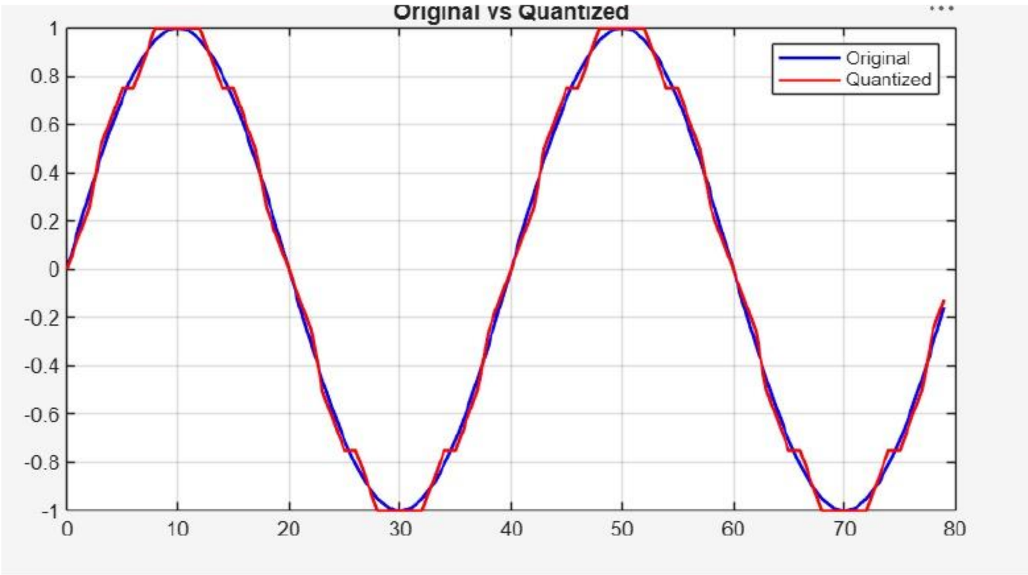
```
clc; clear; close all
n=0:79;
x=sin(2*pi*n/40);
q=8;
xt=floor(x*q)/q;
xr=round(x*q)/q;
et=x-xt;
er=x-xr;
figure
plot(n,x,'b',n,xr,'r','LineWidth',1.5)
grid on
title('Original vs Quantized')
legend('Original','Quantized')
figure
plot(n,x,'b',n,xt,'r',n,xr,'g','LineWidth',1.5)
grid on
title('Truncation vs Rounding')
legend('Original','Truncated','Rounded')
figure
stem(n,et,'r'); hold on; stem(n,er,'g')
grid on
title('Quantization Error')
legend('Truncation','Rounding')
figure
bar([mean(et.^2) mean(er.^2)])
set(gca,'XTickLabel',{'Truncation','Rounding'})
grid on
title('MSE Comparison')
ylabel('MSE')
```

Output / Analysis

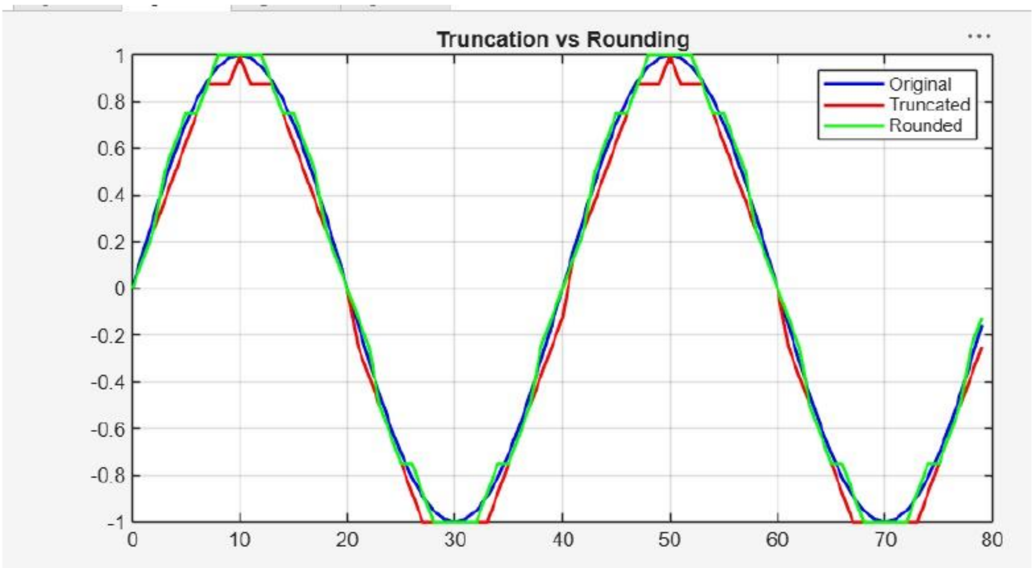
The output/analysis pages below are reproduced from the uploaded official ECA09 MATLAB lab manual for this experiment.

```
et=x-xt;
er=x-xr;
% Fig 1
figure
plot(n,x,'b',n,xr,'r','LineWidth',1.5)
grid on
title('Original vs Quantized')
legend('Original','Quantized')
% Fig 2
figure
plot(n,x,'b',n,xt,'r',n,xr,'g','LineWidth',1.5)
grid on
title('Truncation vs Rounding')
legend('Original','Truncated','Rounded')
% Fig 3
figure
stem(n,et,'r')
hold on
stem(n,er,'g')
grid on
title('Quantization Error')
legend('Truncation','Rounding')
% Fig 4
figure
bar([mean(et.^2) mean(er.^2)])
set(gca,'XTickLabel',{'Truncation','Rounding'})
grid on
title('MSE Comparison')
ylabel('MSE')
```

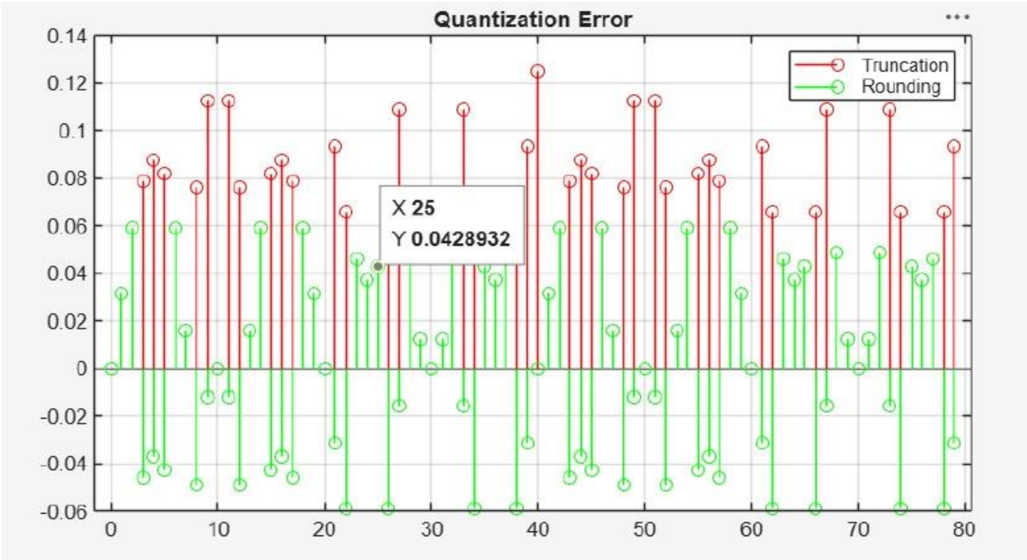
Output:

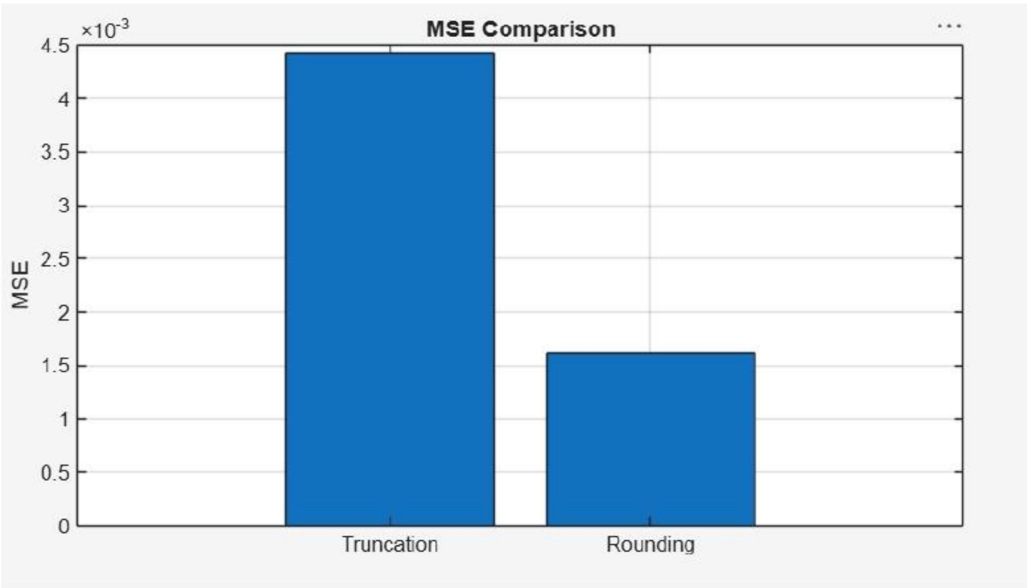


Original vs Quantized
Original
Quantized



Truncation vs Rounding
Original
Truncated
Rounded





Result:

1. The digital signal was successfully represented using a finite word length format in MATLAB.
2. Truncation and rounding operations were implemented on the signal samples.

3. The errors introduced by truncation and rounding were computed and compared, showing that rounding generally produces smaller errors than truncation.
4. The effect of finite precision on signal representation was analyzed, demonstrating that reduced word length increases representation error and may affect signal accuracy.

Practical Question-1:

Generate a discrete-time sinusoidal signal sampled at 2000 Hz and represent it using finite precision arithmetic. Implement truncation and rounding to three decimal places.

Task:

1. Generate a discrete-time sinusoidal signal sampled at **2000 Hz** using MATLAB.
2. Represent the generated signal using **finite precision arithmetic**.
3. Apply **truncation and rounding operations up to three decimal places** on the signal samples.
4. Compute the **error signals and Mean Square Error** (**MSE**) for both truncation and rounding, and compare their performance.

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Practical Question 2:

Analyze the effects of finite word length representation on a discrete-time sinusoidal signal in MATLAB. Consider a sinusoidal signal with a frequency of 50 Hz sampled at 1000 Hz and represent the signal using an 8-bit word length. Implement truncation and rounding operations on the signal samples.

Task:

1. Generate a discrete-time sinusoidal signal in MATLAB with a frequency of 50 Hz sampled at 1000 Hz.
2. Represent the generated signal using an 8-bit finite word length format.
3. Implement truncation and rounding operations on the signal samples.
4. Analyze the effects of finite word length by comparing the original, truncated, and rounded signals in terms of signal distortion and accuracy.

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Result

To compare truncation and rounding effects under finite word length and evaluate their quantization errors and MSE.

Student Details

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